Canonical DE operator contracts and trace identity

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Scope and ETA. A compilable HyMeKo dependency description of the frozen DE/rand/1/bin pilot, structural Query checks with negative cases, canonical operator identity export, and identity binding to existing recorded events. Estimate 40–60 minutes: source/validator, focused tests, recorded-trace binding, documentation. No optimization campaign, simulation, external model call or Bat reconstruction. Structural checks do not prove implementation equivalence.

Ownership. Core's CompiledModel owns compilation, IR and semantic identity; DerivedPathIndex owns canonical paths. Existing QueryEngine predicates, effective fields, exact inheritance and match-count checks own structural query semantics. A Formats research example owns DE-specific validation and JSON export. The local pilot binds exported IDs without parsing HyMeKo. HySTL remains the only temporal monitor and requires no semantic change.

Files. New Formats example examples/metaheuristic_contract.rs and data/metaheuristics/de_rand_1_bin.hym focused tests embedded in that example. The local docs/research/metaheuristics-pilot gains bind_operator_ids.py and test_operator_ids.py; README, PROTOCOL and PROMPT receive precise integration notes. The parent-approved optional prompt step adds build_prompt_inputs.py, extracting an interface stub from the existing reference AST and assembling unsubmitted P/H prompt bundles from the same frozen pseudocode and contract facts; its tests live with the pilot identity checks. No provider is selected or invoked by this builder. This four-file plan and reports/2026-09-29-metaheuristic-contract.md record provenance. Generated contract JSON, DOT/SVG and a bound copy of an existing trace go under the local pilot artifacts. No shared manifests, dependencies, Core, Tensor, Kawamirai, adversarial-grid or existing overview/task files change.

Representation. Explicit typed input/output dependency declarations carry exact canonical state references. Source currently leaves PortDirection unspecified; no direction is inferred from sign. Donor roles, old/new generation state, boundary handling, equality replacement, budgets and random-decision contract remain distinct. Export identity pairs the core semantic digest with the core-derived canonical path; raw DeclIds are snapshot-local diagnostics only.

Validation. Query checks assert vocabulary/profile uniqueness, operator keys, input/output roles, state references and fixed DE profile values. Negative fixtures cover missing/duplicate operator identity, omitted/wrong donor dependency, reversed roles, foreign same-name types and semantic-profile drift. Canonical emission/recompilation must preserve validated meaning and identity. Python checks reject unknown event keys and registry/profile mismatch; they verify recorded numeric data is unchanged. No graph interpreter is implemented.

Budget and risk. One fixed small source, maximum 128 KiB, 4096 declarations and 128 operator roles for the input adapter. Query witnesses capped at eight, no topology expansion or remote fetch by the fixture compiler. Work is source-size bounded; tests target under 30 seconds after cached build. Existing recorded trace binding is capped at 16 MiB input. No measured

speedup or latency claim is made. Canonical paths, not bare numeric IDs, avoid declaration-order aliasing. The dirty Query owner is explicitly hashed; this is a local integration until those changes are published. Duplicate framework revisions are checked with cargo tree -d before edits.

State and rollback. Reuse transformation feat/camera-perception, HEAD 3421b4dabe4a173d52e64ed143db6
Fetch all remotes succeeded; HEAD equals fresh upstream and is seven commits ahead of origin/main. Preserve all existing edits and both worktrees. CORE.YAML items touched: empty. No commit, push, branch switch or new worktree. Rollback only this task's named new files and its pilot documentation additions. Parent handles report archival.

